

LESSON 8.4

AREA OF COMPOSITE FIGURES



LESSON INTRODUCTION

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

LEARNING TARGETS

- I can find the area of composite figures.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.

WORKING TOWARD

MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.

ESSENTIAL QUESTIONS

- How do you find the area of a composite figure?
- How can you use the areas of triangles and rectangles to find the area of a composite figure?

LESSON NARRATIVE

This lesson progresses from refreshing students' prior knowledge of finding the area of a quadrilateral by decomposing it to solving mathematical and real-world problems involving the area of a quadrilateral and composite figures. Students decompose composite figures to find the area.

COMPONENT SUMMARY

8.4.1 WARM-UP (~5 MIN.)

Find the area of a quadrilateral/ parallelogram

8.4.3 GUIDED PRACTICE (~10-15 MIN.)

Decompose composite figures into triangles and rectangles to find their area

8.4.5 WRAP-UP (~5 MIN.)

Students demonstrate their understanding of the lesson using a coordinate plane

8.4.2 GUIDED INSTRUCTION (10-15 MIN.)

Discover multiple ways to decompose a composite figure and find its area

8.4.4 YOUR TURN! (10-15 MIN.)

Find the area of a composite figure using the areas of triangles and rectangles

RESOURCES

GLOSSARY

Key Terms:

- Composite figure
- Area of a composite figure

Additional Terms: N/A

MATERIALS

- Four-function calculator
- (Optional) Separate paper
- (Optional) Square and rectangle manipulatives

ADDITIONAL PREPARATION

- Consider making copies of cutouts of each composite figure to allow students to manipulate the shape.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly split up the figure and get stuck when they do not have a measurement needed for a side.
- Students may think a figure can only be decomposed into two shapes when more than two shapes are needed to find the area.
- Students may not realize they need to use two measurements to find a third measurement that is not given.

THINGS TO CONSIDER

- There may be students who struggle to visualize where to split the composite figure.
- Spend time showing students that there are different ways to divide a composite figure. Encourage students to choose ways that have the most given side lengths.
- Encourage students to write out the formula for the area of a triangle and a rectangle to not confuse the two.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

8.4.3 Guided Practice provides an opportunity for students to engage in a Think-Pair-Share routine. First, students consider the problem. Next, they share their initial thoughts with a partner and solidify their thinking. Finally, students share out their responses with the class.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

Throughout the lesson, the student is analyzing problems in a way that makes sense for the content. The complexity of the diagrams and the multiple entry points build perseverance by challenging the student.

DIFFERENTIATE INSTRUCTION

The intent of Lesson 4 centers on using decomposition to find area, not finding area alone. Students who still struggle finding area should leverage On-Ramp to 6th Grade videos on finding Area. Consider having students preview this lesson by watching the On-Ramp video or have the video available to watch for students to apt-in during 8.4.4 Your Turn! questions.

8.4.1 WARM-UP: BELL WORK

Component Overview: Students will activate prior knowledge by demonstrating understanding from the previous lesson by decomposing a parallelogram to find its area. Students will need to rely on the units on the graph for their measurements.

TEACHER MOVES

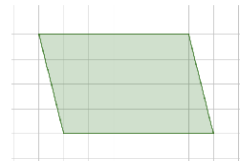
Consider requiring students to draw how they decomposed the quadrilateral and to label the decomposition. Their drawing will help determine common misconceptions students may be having and inform your instruction. Use the series of bell works and ticket out the doors as a picture of each students' progress and as necessary to incorporate into instruction reteaching.



8.4.1 Warm-Up: Bell Work

In the figure, each square of the grid paper is 1 centimeter (cm) by 1 cm.

1. Find the area of the shaded region. Include units.



$$A = b \times h = 6 \text{ cm} \times 4 \text{ cm} = 24 \text{ cm}^2$$

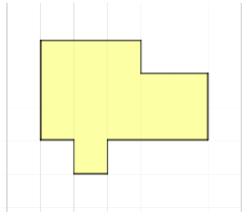


8.4.2 Guided Instruction: Area of Composite Figures



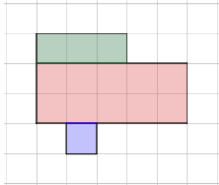
A **composite figure** is a two- or three-dimensional figure that can be decomposed into smaller figures.

Jackson and Danna were instructed to find the area of the composite figure below. The length of each square is 1 centimeter (cm).

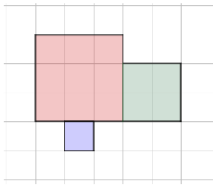


Both Jackson and Danna decomposed the figure, but their teacher noticed they each decomposed it differently.

Jackson



Danna



1. Complete the table about each student's work. Include units.

Student	Area of Red Rectangle	Area of Blue Rectangle	Area of Green Rectangle
Jackson	10 cm ²	1 cm ²	3 cm ²
Danna	9 cm ²	1 cm ²	4 cm ²

2. To find the area of the composite figure, find the sum of the areas of the rectangles for each student. Include units.

	Area of the Composite Figure
Jackson	14 cm ²
Danna	14 cm ²

3. Explain why both students got the same area of the composite shape.

Both students got the same area of the composite shape because it does not matter which way you break down a shape, the area is the same.



The **area of a composite figure** is the sum of the areas of the figure's decomposed parts.

8.4.2 GUIDED INSTRUCTION: AREA OF COMPOSITE FIGURES

Component Overview: Students will explore how to decompose a composite figure into multiple shapes and find their areas, whose sum will be the area of the composite figure. It is important to show students how there are multiple ways a shape can be decomposed and still get the correct area of the figure.

ENRICHMENT

To highlight the multiple ways students can decompose composite figures, ask questions like:

- Is either student correct or incorrect? Why or why not?
- How are Jackson's and Danna's answers the same? How are they different?
- How would you have decomposed the composite shape?
- How could the decomposition help you find the area of the composite figure?

TEACHER MOVES

To further scaffold question 3, ask questions like:

- Does decomposing the composite figure change the shape overall? Why or why not?
- What do you notice about the areas of the smaller shapes? What does this mean for the area of the composite figure?

8.4.3 GUIDED PRACTICE: AREA OF COMPOSITE FIGURES

Component Overview: Students will practice decomposing a composite figure. Students will be given a figure in which they can choose how to break up to find the areas of each shape in the figure to find the total area of the composite figure.

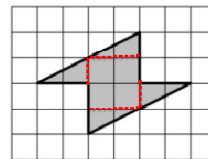
DIFFERENTIATE INSTRUCTION

This activity provides visual entry points through the representations of the shapes shown. Auditory entry points can be provided through partner or whole-class discussion or reading aloud the questions. Consider providing cutouts of the shapes to provide kinesthetic entry points for all learners.



8.4.3 Guided Practice: Area of Composite Figures

1. A composite shape is shown. The length of each square is 1 inch (in).



- Draw lines to decompose the figure into rectangles and/or triangles.
- Find the area of each rectangle and/or triangle. Include units.

$1 \text{ in}^2, 1 \text{ in}^2, 1 \text{ in}^2, 1 \text{ in}^2, 4 \text{ in}^2$

- Find the sum of the areas of the rectangles and/or triangles to find the area of the composite figure. Include units.

8 in^2

- Find a classmate who decomposed the shape differently than you did. Describe their process.
Answers will vary. My classmate divided the shape into two triangles. My classmate divided the shape into 3 triangles and 1 rectangle.

TEACHER MOVES

Think-Pair-Share

Before answering question 2, engage students in a Think-Pair-Share routine. Ask guiding questions to provide further scaffolding like:

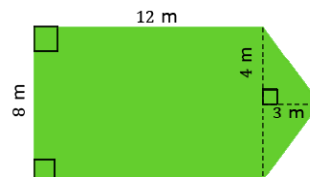
- How many shapes can you decompose this composite figure into?
- How can you use those shapes to find the area of the composite figure?

Have students first share their responses with a partner. Then share out as a class and clarify any misconceptions.

DIFFERENTIATE INSTRUCTION

Consider prompting students to utilize a manipulative like a piece of paper for replicating this decomposition.

2. Thomas has decided to put grass sod in his backyard. A drawing of his backyard is shown.



How much sod does Thomas need to buy?

$$A_{\text{rectangle}} = 8\text{m} \times 12\text{m} = 96 \text{ m}^2$$

$$A_{\text{triangle}} = \frac{8\text{m} \times 3\text{m}}{2} = 12 \text{ m}^2$$

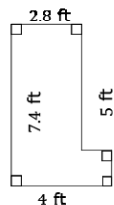
$$A_{\text{total}} = A_{\text{rectangle}} + A_{\text{triangle}} = 96\text{m}^2 + 12\text{m}^2 = 108 \text{ m}^2$$



8.4.4 Your Turn!

For each composite figure, find the area, in square feet (ft^2). Show your work.

1.



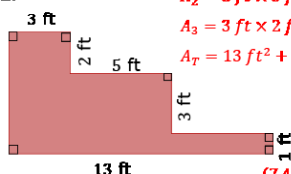
$$A_1 = 4 \text{ ft} \times (7.4 \text{ ft} - 5 \text{ ft}) = 9.6 \text{ ft}^2$$

$$A_2 = 5 \text{ ft} \times 2.8 \text{ ft} = 14 \text{ ft}^2$$

$$A_T = A_1 + A_2 = 9.6 \text{ ft}^2 + 14 \text{ ft}^2 = 23.6 \text{ ft}^2$$

$$\text{Area} = \underline{23.6 \text{ ft}^2}$$

2.



$$A_1 = 13 \text{ ft} \times 1 \text{ ft} = 13 \text{ ft}^2$$

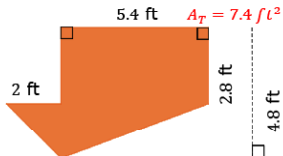
$$A_2 = 8 \text{ ft} \times 3 \text{ ft} = 24 \text{ ft}^2$$

$$A_3 = 3 \text{ ft} \times 2 \text{ ft} = 6 \text{ ft}^2$$

$$A_T = 13 \text{ ft}^2 + 24 \text{ ft}^2 + 6 \text{ ft}^2 = 43 \text{ ft}^2$$

$$\text{Area} = \underline{43 \text{ ft}^2}$$

3.



$$A_1 = \frac{(7.4 \text{ ft}) \times (2 \text{ ft})}{2} = 7.4 \text{ ft}^2$$

$$A_2 = 5.4 \text{ ft} \times 2.8 \text{ ft} = 15.12 \text{ ft}^2$$

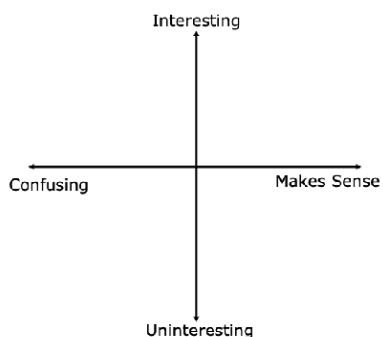
$$A_T = 7.4 \text{ ft}^2 + 15.12 \text{ ft}^2 = 22.52 \text{ ft}^2$$

$$\text{Area} = \underline{22.52 \text{ ft}^2}$$



8.4.5 Wrap-Up: Reflection

- Plot a point in one of the quadrants below to indicate how you feel about today's learning target: "I can find the area of composite figures."



Answers will vary.

8.4.4 YOUR TURN!

Component Overview: Students will demonstrate understanding of decomposing a composite figure to find the area of the shapes that make up that figure in order to find the total area using the given measurements.

DIFFERENTIATE INSTRUCTION

Consider differentiating by content with one of the following strategies:

- Choice: Allow students to choose two out of three questions to complete.
- Support: Prompt students to watch the Study Expert video from 8.4.3 Guided Practice for a model or exemplar.
- Gridlines: Provide students with graph paper to trace each composite figure, then use the gridlines to better make sense of ways to decompose each figure.

QUESTIONING

Challenge students who finish early to create their own composite figure, including dimensions with at least one missing dimension, to trade with a partner and submit for extra credit.

8.4.5 WRAP-UP: REFLECTION

Component Overview: Students will demonstrate their understanding of finding the area of composite figures in a coordinate-plane format. Encourage students to answer truthfully to better reinforce concepts the following day. Consider using a graph on the wall to have all students graph their answer.

DIFFERENTIATE INSTRUCTION

Consider prompting students to include an ordered pair with their response in addition to rationale as to why they chose that specific location.

For example, "I chose (4,3) because it made sense and I had found the shapes interesting."